Supergeometry of Gauged p-Form Hierarchies

(Based on work with M. Becker, K. Becker, and D. Robbins)

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Motivation: Eleven-dimensional Supergravity

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Action ($G = dC$):

$$S \sim S_{EH} + \frac{i}{2}\int d^{11}x\, \bar{\psi}_m \Gamma^{mnp} \nabla_n \psi_p + \frac{1}{2}\int G \wedge * G + \int C \wedge G \wedge G$$

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2. Supersymmetry algebra does not close on gravitino unless we impose $\Gamma^{mnp}\nabla_n\psi_p = 0$

mixed *kinematical* (first class/off-shell) and *dynamical* (second class/on-shell)
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Super(symmetric)space does not help because we need

1. Supersymmetric space as Klein geometry G/H with $G =$ eleven-dimensional super-Poincaré group and $H = Spin(10,1)$ Lorentz subgroup.
2. Maximally non-integrable odd distribution $\{D\}$: $[D, D] = -i\, \not{\partial}$ and $\not{\partial}^2 = \Box$
3. "Large enough" integrable odd sub-distribution $\{\bar{D}\}$ ($\not{\sharp}$ in 11D)

Goal

Suggested solution:

1. Reduce $H \to H_{st} \times H_{int}$ (and $G \to G_{st} \times G_{int}$) such that Klein geometry (G_{st}, H_{st}) admits integrable odd sub-distribution
2. Replace 11D superspace by foliation by G_{st}/H_{st} superspaces

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- D > 4 and/or $N > 1$ supergravity contains graviton and gravitino but also p-form fields + Chern-Simons-like terms (*e.g.* 5D, $N = 1 \Rightarrow p = 1$ and 11D, $N = 1 \Rightarrow p = 3$)

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- Superforms formulated in terms of frames
  $\Rightarrow$ Reduction to 4D gives Kaluza-Klein gauge fields ${e_m}^i$
  $\Rightarrow$ non-abelian gauging of p-form hierarchy

$N = 0$

- Spacetime M with p-form gauge field $\phi_{[p]}$ transforming under abelian gauge transformation as $\delta\phi_{[p]} = d_M \Lambda_{[p-1]}$, with invariant field strength $F_{[p+1]} = d_M \phi_{[p]}$, and Chern-Simons-like invariant

$$S_{CS} = \int_M \phi \wedge d\phi \wedge \cdots \wedge d\phi = \int_M \phi \wedge c$$

- Splitting $M = X \times Y \Rightarrow d_M = d_X + d_Y$ and $\Omega^p_M = \bigoplus_{q=0}^p \Omega^{p-q}_X(\Omega^q_Y)$ double complex.
- Write $d := d_X$ and $\partial := d_Y$. Forms split $\phi_{[p]} \to \sum_q \phi_{[p-q,q]}$ by bi-degree and

$$\delta\phi_{[r,s]} = d\Lambda_{[r-1,s]} + \partial\Lambda_{[r,s-1]} \quad \text{and} \quad F_{[r,s]} = d\phi_{[r-1,s]} + \partial\phi_{[r,s-1]}$$

- Chern-Simons-like action for $\{\phi_{[r,s]}\}_{r+s=p}$:

$$S_{CS} = \sum_{r=0}^{p} \int_{X\times Y} \phi_{[r,p-r]} \wedge c_{[\dim(X)-r,\dim(Y)-p+r]}$$

- Invariant (up to surface terms) provided c is a closed: $dc_{[r,s]} + (-1)^p \partial c_{[r+1,s-1]} = 0$

4D, $N = 1$ Superspace

- "Physical" superspace: Klein geometry $X = G/H$ with $G =$ super-Poincaré (4D, $N = 1$) and $H \subset G$ Lorentz subgroup $Spin(3,1) \cong SL_2(\mathbf{C})$

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- Left-invariant framing of the tangent space $[Q, D] = 0$:

$$D_A = e_A{}^M \partial/\partial z^M = \begin{cases} D_a = \partial_a := \delta_a^m \partial/\partial x^m \\ D_\alpha = \delta_\alpha^\mu \partial/\partial\theta^\mu - i(\sigma^m)_{\alpha\dot{\alpha}} \delta_{\dot{\mu}}^{\dot{\alpha}} \bar{\theta}^{\dot{\mu}} \partial/\partial x^m \\ \bar{D}_{\dot{\alpha}} = \delta_{\dot{\alpha}}^{\dot{\mu}} \partial/\partial\theta^\mu - i(\sigma^m)_{\alpha\dot{\alpha}} \delta_\mu^\alpha \theta^\mu \partial/\partial x^m \end{cases}$$

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$$[D, D] = 0 \quad , \quad [D, \bar{D}] = -2i\partial \quad , \quad [\bar{D}, \bar{D}] = 0$$

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  $\Rightarrow$ off-shell, irreducible superfield representations for any superspin

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- Super-de Rham differential $d = dz^M \partial/\partial z^M \to e^A D_A$ but co-frame carries torsion $de^A = T^A$ (even in flat superspace) $\Rightarrow$ complicates analysis of closed super-p-forms

Closed Superforms

Solving $dF = 0$ in superspace (a.k.a. solving the "Bianchi identities") gives

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\begin{array}{ll}
p = 0 & F = f \\
p = 1 & F = e^\alpha D_\alpha F + e^a (D\sigma_a \bar{D}) F \\
p = 2 & F = e^\alpha e^a (\sigma_a \bar{W})_\alpha + e^a e^b (D\sigma_{ab} W) + \text{h.c.} \\
p = 3 & F = e^\alpha e^{\dot{\alpha}} e^a (\sigma_a)_{\alpha\dot{\alpha}} H + \cdots + e^a e^b e^c \epsilon_{abcd} (D\sigma^d \bar{D}) H \\
p = 4 & F = e^\alpha e^\beta e^a e^b (\sigma_{ab})_{\alpha\beta} G + \cdots + e^a e^b e^c e^d \epsilon_{abcd} D^2 G
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together with the conditions on the superfield strengths

p	BI	irreducibility		solution	prepotential
0	$\partial f = 0$	$\bar{f} = -f$	$\Rightarrow$	$f = ic$	$c \in \mathbf{R}$
1	$\bar{D}^2 DF = 0$	$\bar{F} = F$	$\Rightarrow$	$F = \frac{1}{2i}\left(\Phi - \bar{\Phi}\right)$	$\bar{D}\Phi = 0$
2	$DW - \bar{D}\bar{W} = 0$	$\bar{D}W = 0$	$\Rightarrow$	$W = -\frac{1}{4}\bar{D}^2 DV$	$\bar{V} = V$
3	$\bar{D}^2 H = 0$	$\bar{H} = H$	$\Rightarrow$	$H = \frac{1}{2i}\left(D\Sigma - \bar{D}\bar{\Sigma}\right)$	$\bar{D}\Sigma = 0$
4		$\bar{D}G = 0$	$\Rightarrow$	$G = -\frac{1}{4}\bar{D}^2 X$	$\bar{X} = X$
"5"				Γ	$\bar{D}\Gamma = 0$

Prepotential de Rham Complex

Organize into (exact) cochain complex of sheaves of superfields

$$0 \longrightarrow i\mathbf{R} \longrightarrow \underbrace{\mathscr{O}^0}_{\text{chiral scalar}} \xrightarrow{d_0} \underbrace{\mathscr{O}^1}_{\text{real scalar}} \xrightarrow{d_1} \underbrace{\mathscr{O}^2}_{\text{chiral spinor}} \xrightarrow{d_2} \underbrace{\mathscr{O}^3}_{\text{real scalar}} \xrightarrow{d_3} \underbrace{\mathscr{O}^4}_{\text{chiral scalar}} \xrightarrow{d_4} 0$$

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For $\mathbf{A} = (\Phi, V, \Sigma, X, \Gamma) \in \mathscr{O}^\bullet$ the poly-superform $\mathbf{F} = d\mathbf{A} = (0, F, W, H, G)$ is invariant under $\delta\mathbf{A} = d\mathbf{\Lambda}$.

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NB: These are not derivations: $d_i(\phi\phi') \neq (d_i\phi)\phi' + \phi d_i\phi'$

There is no differential graded algebra structure at the level of prepotentials.

Simple Superspace Foliation

Take spacetime $M = X \times Y$ with $X = \mathbf{R}^{4|4}$ and Y an ordinary $N = 0$ space.
Extend all $\mathscr{O}_X^\bullet$ to take values in the de Rham complex $(\Omega_Y^\bullet, d_Y)$; denote $d_Y = \partial$

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Example: 11D SG has $Y = \mathbf{R}^7$ and 3-form decomposing under $\Omega_M^3 \to \bigoplus_{q=0}^3 \Omega_X^{3-q}(\Omega_Y^q)$:

$$
\begin{array}{cccccccccc}
 & 1 & & 7 & & \binom{7}{2} = 21 & & \binom{7}{3} = 35 \\
 & \text{3-form} & & \text{2-forms} & & \text{vectors} & & \text{scalars} \\
C_{[3]} \longrightarrow & C_{[3,0]} & + & C_{[2,1]} & + & C_{[1,2]} & + & C_{[0,3]} \\
 & X_{\mathbf{1}} & & \Sigma_{\mathbf{7}} & & V_{\mathbf{21}} & & \Phi_{\mathbf{35}} \\
F_{[4]} \longrightarrow & F_{[4,0]} & + & F_{[3,1]} & + & F_{[2,2]} & + & F_{[1,3]} & + & F_{[0,4]} \\
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Field strengths and Bianchi identities (relative closure relations):

$$
\begin{array}{rcrcl}
E & = & & - & \partial\Phi \\
F & = & d_0(\Phi) & - & \partial V \\
W & = & d_1(V) & - & \partial\Sigma \\
H & = & d_2(\Sigma) & - & \partial X \\
G & = & d_3(X) & &
\end{array}
\quad \Longrightarrow \quad
\begin{array}{rcl}
d_0(E) & = & -\partial F \\
d_1(F) & = & -\partial W \\
d_2(W) & = & -\partial H \\
d_3(H) & = & -\partial G \\
d_4(G) & = & 0
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Take spacetime $M = X \times Y$ with $X = \mathbf{R}^{4|4}$ and Y an ordinary $N = 0$ space.
Extend all $\mathscr{O}_X^\bullet$ to take values in the de Rham complex $(\Omega_Y^\bullet, d_Y)$; denote $d_Y = \partial$
Example: 11D SG has $Y = \mathbf{R}^7$ and 3-form decomposing under $\Omega_M^3 \to \bigoplus_{q=0}^3 \Omega_X^{3-q}(\Omega_Y^q)$:

$$
\begin{array}{lcccccccc}
 & 1 & & 7 & & \binom{7}{2} = 21 & & \binom{7}{3} = 35 \\
 & \text{3-form} & & \text{2-forms} & & \text{vectors} & & \text{scalars} \\
C_{[3]} \longrightarrow & C_{[3,0]} & + & C_{[2,1]} & + & C_{[1,2]} & + & C_{[0,3]} \\
 & X_{\mathbf{1}} & & \Sigma_{\mathbf{7}} & & V_{\mathbf{21}} & & \Phi_{\mathbf{35}} \\
F_{[4]} \longrightarrow & F_{[4,0]} & + & F_{[3,1]} & + & F_{[2,2]} & + & F_{[1,3]} \quad + \quad F_{[0,4]} \\
 & G_{\mathbf{1}} & & H_{\mathbf{7}} & & W_{\mathbf{21}} & & F_{\mathbf{35}} \quad\quad E_{\mathbf{35}}
\end{array}
$$

Field strengths and Bianchi identities (relative closure relations):

$$
\begin{array}{lcll}
E & = & & - \ \partial\Phi \\
F & = & d_0(\Phi) & - \ \partial V \\
W & = & d_1(V) & - \ \partial\Sigma \\
H & = & d_2(\Sigma) & - \ \partial X \\
G & = & d_3(X) &
\end{array}
\quad \Longrightarrow \quad
\begin{array}{lcl}
d_0(E) & = & -\partial F \\
d_1(F) & = & -\partial W \\
d_2(W) & = & -\partial H \\
d_3(H) & = & -\partial G \\
d_4(G) & = & 0
\end{array}
$$

Manifestly supersymmetric "Abelian Tensor Hierarchy"

Abelian Chern-Simons-like Action

"Abelian Tensor Hierarchy" has Chern-Simons-like invariants of the form (prepotential) $\times$ (superfield strength) $\times \cdots \times$ (superfield strength)

$$S_{CS} = \int d^4x \int_Y \left\{ \int d^2\theta \left[\Phi \wedge c_{[4,4]} + \Sigma^\alpha \wedge c_{[2,6]\alpha} \right] + \text{h.c.} \right.$$
$$\left. + \int d^4\theta \left[V \wedge c_{[3,5]} + X \wedge c_{[1,7]} \right] \right\}$$

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Gauge invariant (up to surface terms) under

$$\delta(\Phi, V, \Sigma, X) = (0, d_0\Lambda, d_1 U, d_2 \Upsilon) + \partial(\Lambda, U, \Upsilon, 0)$$

provided the cs satisfy the "descent relations" (cocycle condition)

$$\begin{array}{ccc} 0 & = & \partial c_{[1,7]} \\ d_1(c_{[1,7]}) & = & \partial c_{[2,6]} \\ d_2(c_{[2,6]}) & = & \partial c_{[3,5]} \\ d_3(c_{[3,5]}) & = & \partial c_{[4,4]} \\ d_4(c_{[4,4]}) & = & 0 \end{array}$$

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Formally the same equations as the Bianchi identities...
but far less trivial because c's are bilinear in field strengths and d_i are not derivations

Existence of Quadratic Co-cycle

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Here $\omega(W, F)$ is the "Chern-Simons superform"

$$\omega(W, F) := W^\alpha \wedge D_\alpha F + \tfrac{1}{2} D^\alpha W_\alpha \wedge F + \text{h.c.}$$

It is the superspace analogue of $A \wedge F$ satisfying

$$\begin{aligned}
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Using every D-identity, Bianchi identity, and internal de Rham DGA property, it is possible to show that $c \in \bigoplus_i \mathscr{O}^{8-i}(\Omega^i_Y)$ is a cocycle.

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Important example: Let Y be a smooth manifold and set $(\mathfrak{g}, f) = (\mathfrak{X}(Y), [\cdot,\cdot])$ to be the algebra of diffeomorphisms and $(V_\bullet, q) = (\Omega^\bullet_Y, d_Y)$ to be the de Rham complex. Finally, for any vector field $a \in \mathfrak{X}(Y)$, set $t_a := \mathscr{L}_a$ to be the Lie derivative along a and set $h_a := \iota_a$ to be the contraction of forms by the vector field.

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- Repeat analysis of abelian super-p-form hierarchy: For $\mathbf{A} \in \boldsymbol{\Omega}^1$, $\boldsymbol{\Lambda} \in \boldsymbol{\Omega}^0$, *et cetera*

$$\delta\mathbf{A} = t_\lambda \mathbf{A} + \boldsymbol{d}\boldsymbol{\Lambda} \quad , \quad \mathbf{F} = \boldsymbol{d}\mathbf{A} \quad \Rightarrow \quad \delta\mathbf{F} = t_\lambda \mathbf{F} \quad \text{and} \quad \boldsymbol{d}\mathbf{F} = 0.$$

Non-Abelian Result

Returning to the 11D supergravity example (*e.g.* $q \sim \partial$)

$$\begin{aligned} E &= -\partial\Phi \\ F &= d_0(\Phi) - \partial V \\ W &= d_1(V) - \partial\Sigma - \iota_{\mathscr{W}}\Phi \\ H &= d_2(\Sigma) - \partial X - \omega(\mathscr{W}, V) \\ G &= d_3(X) - \iota_{\mathscr{W}}\Sigma \end{aligned} \implies \begin{aligned} 0 &= -\partial E \\ d_0(E) &= -\partial F \\ d_1(F) &= -\partial W - \iota_{\mathscr{W}} E \\ d_2(W) &= -\partial H - \omega(\mathscr{W}, F) \\ d_3(H) &= -\partial G - \iota_{\mathscr{W}} W \\ d_4(G) &= 0 \end{aligned}$$

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The field strengths are all gauge covariant and satisfy the modified Bianchi identities. The modified quadratic forms

$$\begin{aligned} \mathscr{O}^1(\Omega^7_Y) \ni c_{[1,7]} &= \mathrm{Re}(E) \wedge F \\ \mathscr{O}^2(\Omega^6_Y) \ni c_{[2,6]} &= -E \wedge W - \tfrac{i}{4}\bar{\mathscr{D}}^2(F \wedge \mathscr{D}F) \\ \mathscr{O}^3(\Omega^5_Y) \ni c_{[3,5]} &= -\mathrm{Re}(E) \wedge H - 2\omega(W, F) - i\iota_{\mathscr{W}}(F \wedge \mathscr{D}F) + i\iota_{\bar{\mathscr{W}}}(F \wedge \bar{\mathscr{D}}F) \\ \mathscr{O}^4(\Omega^4_Y) \ni c_{[4,4]} &= EG + 2W \wedge W - \tfrac{i}{2}\bar{\mathscr{D}}^2(FH) \end{aligned}$$

miraculously satisfy the modified descent relations.

Non-Abelian Chern-Simons Action

To write an invariant action we need the analogue of a trace on the non-abelian p-form hierarchy structure. In the compactification case, this is $\int_Y$ with the conditions

$$\int_Y \partial(\dots) = 0 \quad \text{and} \quad \int_Y \iota_\lambda(\dots) = 0 \quad \Rightarrow \quad \int_Y \mathscr{L}_\lambda(\dots) = 0$$

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In a more general case, we need a 3-chain α on the non-abelian p-form hierarchy closed under the differentials q and $h_\bullet$ (and therefore gauge-invariant). Explicitly, we need a chain

$$\alpha : \bigoplus_{r+s+t=4} \mathbf{\Omega}^r \otimes \mathbf{\Omega}^s \otimes \mathbf{\Omega}^t \to \mathbf{R}$$

subject to

$$\begin{aligned} q\alpha(\mathbf{x},\mathbf{y},\mathbf{z}) &:= \alpha(q\mathbf{a},\mathbf{b},\mathbf{c}) + \alpha(\mathbf{a},q\mathbf{b},\mathbf{c}) + \alpha(\mathbf{a},\mathbf{b},q\mathbf{c}) = 0 \\ h_a\alpha(\mathbf{x},\mathbf{y},\mathbf{z}) &:= \alpha(h_a\mathbf{x},\mathbf{y},\mathbf{z}) + \alpha(\mathbf{x},h_a\mathbf{y},\mathbf{z}) + \alpha(\mathbf{x},\mathbf{y},h_a\mathbf{z}) = 0 \quad \forall a \in \mathfrak{g} \end{aligned}$$

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- 4D, $N = 1$ component supermultiplets:

$$
\begin{array}{ccl}
1 & : & (e^a_m, \psi^\alpha_m) \oplus (g_{\mathbf{1}}, C_{\mathbf{1}}, C_{mnp}, \chi^\alpha_{\mathbf{1}}) \\
7 & : & (C_{\mathbf{7}m}, \psi^\alpha_{\mathbf{7}m}) \oplus (e_{\mathbf{7}m}, \eta^\alpha_{\mathbf{7}}) \oplus (C_{\mathbf{7}}, C_{\mathbf{7}mn}, \chi^\alpha_{\mathbf{7}}) \\
14 & : & (C_{\mathbf{14}m}, \chi^\alpha_{\mathbf{14}}) \\
27 & : & (C_{\mathbf{27}} + i g_{\mathbf{27}}, \chi^\alpha_{\mathbf{27}})
\end{array}
$$

Faithful Superspace Embedding

- 1 superspin-$\frac{3}{2}$ superconformal graviton $H^a \sim \cdots + (\theta\sigma^m\bar{\theta})e^a_m + \bar{\theta}^2(\theta\sigma^m\psi_m) + \ldots$
- 7 superspin-1 gravitino superfields $\Psi^{\alpha i} \sim \cdots + (\sigma^m\bar{\theta})^\alpha C_{\mathbf{7}m} + (\theta\sigma^m\bar{\theta})\psi^{\alpha i}_m + \ldots$
- 1 4-form "radion" field strength $G_{\mathbf{1}} = (g_{\mathbf{1}} + iC_{\mathbf{1}}) + \theta\chi_{\mathbf{1}} + \theta^2\varepsilon^{mnpq}F_{mnpq} + \ldots$
- 7 3-form field strengths $H_{\mathbf{7}} = C_{\mathbf{7}} + \theta\chi_{\mathbf{7}} + (\theta\sigma_m\bar{\theta})\varepsilon^{mnpq}H_{npq}$
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Open questions and current & future work

Loads of work to do...

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3. Relation to exceptional generalized geometry
4. Investigation of linearized action and resolution of "the **7**-problem"
5. Construction of non-linear superspace action, perhaps in gravitino expansion
6. All orders in α' results for Kähler and super-potential
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THANK YOU!